

HBS formation as a second-order phase transition

A Landau functional for SCIANTIX

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SCIANTIX status

The model

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SCIANTIX status

SCIANTIX had three formation models for the restructured volume fraction

option	form	driven by
1	KJMA, Barani (2020) ¹	effective burnup
2	KJMA + $bu_{inc} = 15$, Biswas and Aagesen (2025) ²	effective burnup
3	KJMA on ρ_d , Veshchunov ³ + Zullo (2026)	local burnup

All three are **transformation kinetics**: a sigmoid fitted in burnup. None of them produces the **mean misorientation** Θ of the subgrains, or the **subgrain radius** r_n .

¹ Barani et al., *J. Nucl. Mater.* 539 (2020) 152296.

² Biswas and Aagesen, *Comput. Mater. Sci.* 258 (2025) 114052, Eq. 45.

³ Veshchunov and Shestak, *J. Nucl. Mater.* 384 (2009) 12–18.

The data this model is built on

paper	technique	Θ	X	r_n
Zacharie-Aubrun (2022), standard UO_2	EBSD	27	27	14
Zacharie-Aubrun (2022), Cr-doped	EBSD	11	11	8
Onofri (2025)	EBSD, no restructuring ($\Theta = 0$)	14	–	–
Gerczak (2018)	image analysis (Barani 2020), $d_A/2$	–	5	4
Noirot (2015)	Xe-depleted area	–	6	–
total		52	49	26

$$\Theta = [A_{\text{Mis}}(f_1 - f_{10}) + 10f_{10}]/100;$$

$$X = f_{10}, \text{ area fraction above } 10^\circ;$$

$$r_n = \text{ECD50 \%}/2.$$

1° is the lower bin edge of the EBSD data.

What KJMA is, and what it assumes

Nuclei appear *at random* in the untransformed material and grow *isotropically*; count the transformed volume as if the growing regions could overlap freely, the **extended volume** V_e , and the overlap correction is statistical:

$$dX = (1 - X) \frac{dV_e}{V} \implies X = 1 - \exp(-K t^n)$$

What you get

K and n , and a sigmoid. n carries the dimensionality of growth and the nucleation regime.

What it assumes

Randomly placed nuclei, a growth law independent of where you are in the transformation, and a *time dependence*.

The time dependence in the KJMA model

- 1 **Effective burnup**, $n = 3.54$, $K = 2.77 \times 10^{-7}$, fitted on restructured-fraction data.
- 2 Same, with an **incubation burnup** $bu_{inc} = 15 \text{ MWd/kgHM}$ below which nothing happens. Biswas & Aagesen get it from a balance between the energy stored in dislocations and the energy of forming a subgrain used to shift the origin of an empirical curve rather than to build the curve.
- 3 KJMA not on burnup but on the **dislocation density** $\rho_d(bu, T)$, with a threshold $\rho_{crit} = 6 \times 10^{14} \text{ m}^{-2}$. The first to make the driver a **microstructural state variable** and the only one that feels temperature.

What none of them can do

They are calibrated on X alone. There is no misorientation angle neither subgrain size.

Why a phase transition could be the right shape for this problem

The HBS forms *continuously*

Not by nucleating new grains, but by low-angle boundaries absorbing dislocations until their

misorientation drifts through the

10° HAGB threshold. In the

metallurgical literature this is

continuous dynamic

recrystallization, and it is a

well-posed alternative to the

nucleation-and-growth picture

KJMA describes.⁵

A continuous transformation has an order parameter

If a single continuous quantity distinguishes the two states,

Landau's argument applies: write the energy as a function of it and keep only what symmetry allows.⁶

$$F(\eta) = \rho_{\text{free}} A_1 G b^2 + \rho_{\text{ord}}(\eta) A_2(\eta) G b^2$$

F depends on η only through η^2 and $|\eta|$: the energy of a boundary cannot depend on the *sign* of the misorientation across it.

F has no threshold of its own: every length in it scales with the density. The threshold comes from the critical density ρ_{crit} in ρ_{tot} .

⁵Gourdet & Montheillet, *Acta Mater.* 51 (2003) 2685; Humphreys, Rohrer & Rollett (2017), chs. 6 and 10.

⁶Landau (1937); Landau & Lifshitz, *Statistical Physics I*, §143.

Landau against KJMA, side by side

	KJMA (options 1–3)	Landau (option 4)
what is solved	a <i>kinetic</i> law along a path	the <i>equilibrium</i> of a state
driver	burnup or ρ threshold	$\rho_{\text{tot}}(bu) - \rho_{\text{crit}}$
free parameters	K , n fitted on X	β , k , ρ_{crit} fitted jointly on Θ , X , r_n
predicts X	by construction	yes , lever rule, no extra parameter
predicts Θ , r_n	no	yes
reversible?	no, by construction	yes, see below

The price of using an equilibrium theory

A drop in ρ_{tot} would un-restructure the fuel. Restructuring is irreversible, so the implementation carries a **monotonic lock** on Θ . However, this is easy if one considers the dislocation density generated by irradiation, and eventually includes the annealing phenomena within the evolution of the Landau free energy.

The model

HBS as a second-order phase transition

Order parameter

$$\eta = \frac{\theta}{\theta_{\max}}, \quad \theta_{\max} \equiv \theta_{\text{HAGB}} = 10^\circ$$

the mean misorientation normalised to the LAGB/HAGB boundary.

External condition: the local burnup, entering *only* through the dislocation density available to polygonize

$$\rho_{\text{tot}}(bu) = \max(10^{2.2 \times 10^{-2} bu + 13.8} - \rho_{\text{crit}}, 0)$$

Nogita & Une (1994)

Free energy

$$F(\eta) = \rho_{\text{free}} A_1 G b^2 + \rho_{\text{ord}} A_2 G b^2$$

Even in η : the energy cannot depend on the *sign* of a misorientation.

Minimising F gives Θ ; the same wall geometry gives r_n ; the lever rule gives X .

Three adjustable parameters

β (wall geometry), k (sweeping), ρ_{crit} (critical density, the density scale F lacks).

The geometric wall

A low-angle boundary *is* a wall of dislocations.

1. Spacing sets the angle $\theta = b/d$

so a wall of n dislocation families carries a line length per unit *wall area*

$$L_A = n\theta/b$$

2. Wall area per unit volume or low angle grain boundary surface

$$S/V = 3\sqrt{\rho_{\text{LAGB}}} / \beta$$

β : the one geometric parameter, of the same kind as Rest & Hofman (2000) but written on ρ_{LAGB} , not on the total network density. $n = 2$, Gourdet & Montheillet, not fitted (between 1-3).

Close the two on each other:

$$\rho_{\text{LAGB}} = L_A \cdot S/V$$

$$\Rightarrow \sqrt{\rho_{\text{LAGB}}} = \frac{3n\theta}{\beta b}$$

Everything follows from θ alone

$$\rho_{\text{LAGB}} = \left(\frac{3n\theta}{\beta b}\right)^2, \quad r_n = \min\left(\frac{1.5}{(S/V) e^{-x}}, R_{\text{grain}}\right)$$

with $S/V = 9n\theta/(\beta^2 b)$ and x the extended swept volume (next slide).

The dislocation balance

Every dislocation is in exactly one of three states, and they must **add up to ρ_{tot}** . With the extended swept volume $x = k \rho_{\text{ord}} / \rho_{\text{tot}}$ (Gourdet & Montheillet: $d\rho_i = -\rho_i dV$, integrated):

$$\underbrace{\rho_{\text{ord}} = \rho_{\text{LAGB}}^{\text{max}} \eta^2}_{\text{condensed into low-angle walls}} + \underbrace{(\rho_{\text{tot}} - \rho_{\text{ord}})(1 - e^{-x})}_{\rho_{\text{swept}}: \text{ annihilated by moving boundaries}} + \underbrace{(\rho_{\text{tot}} - \rho_{\text{ord}}) e^{-x}}_{\rho_{\text{free}}: \text{ still a random array}} = \rho_{\text{tot}}$$

Each population carries the line energy of the state it is in:

$$F = \rho_{\text{free}} A_1 G b^2 + \rho_{\text{ord}} A_2 G b^2, \quad A_i = \frac{f(\nu)}{4\pi} \ln \frac{R_i}{b}$$

$$A_1 = \frac{f(\nu)}{4\pi} \ln \frac{\rho_{\text{tot}}^{-1/2}}{b}, \quad A_2(\eta) = \frac{f(\nu)}{4\pi} \ln \frac{\min(b/\theta, \rho_{\text{tot}}^{-1/2})}{b}$$

the swept ones carry nothing. Here x is a *volume*: a linear $\Delta R/R$ form would differ ($x \approx 0.5$ at 60 GWd/tU, $e^{-x} = 0.6$ against $1 - x = 0.5$).

The functional, collected

$$F = C_0 + E_{\text{wall}} + E_{\text{sweep}}, \quad C_0 = \rho_{\text{tot}} A_1 G b^2$$
$$E_{\text{wall}} = \underbrace{\rho_{\text{ord}} (A_2 - A_1) G b^2}_{\leq 0: \text{ the screening gain}}, \quad E_{\text{sweep}} = \underbrace{-\rho_{\text{swept}} A_1 G b^2}_{\leq 0: \text{ annihilation}}$$

The engine of the transition is $A_2 - A_1$. As burnup rises, ρ_{tot} rises, the screening length due to wall formation shrinks, A_2 falls below A_1 , and it becomes cheaper to store dislocations in walls than to leave them random.

Equilibrium: minimum of F on $[0, \min(\eta_{\text{bal}}, 1)]$, by a coarse scan and a golden section (A_2 carries $\theta \ln \theta$ and the sweep an exponential: no closed form). Both terms are negative from $\eta = 0$ on, so the threshold is ρ_{crit} in ρ_{tot} , and it is continuous.

Condition on the minimisation problem: dislocation balance

The walls cannot hold more dislocations than exist, so $\rho_{\text{ord}} \leq \rho_{\text{tot}}$. The equilibrium is the minimum of F on the **admissible interval**, not the free stationary point:

Eq. (7b) — admissibility

$$\eta_{\text{bal}} = \sqrt{\min\left(\frac{\rho_{\text{tot}}}{\rho_{\text{LAGB}}^{\text{max}}}, 1\right)}, \quad \eta_{\text{eq}} = \underset{[0, \min(\eta_{\text{bal}}, 1)]}{\operatorname{argmin}} F$$

With the current calibration the bound **binds at high burnup, from 89 to 110 GWd/tU (600 K)**, where every dislocation is in a wall and $\Theta = \beta b \sqrt{\rho_{\text{tot}}} / 3n$. Below that the minimum is interior.

Calibration

The objective and the three data sets

Joint fit of β, k, ρ_c ($n = 2$ fixed) on all three observables:

$$J = \frac{\langle \Delta \Theta^2 \rangle_w}{\text{var } \Theta} + w_r \frac{\langle \Delta r^2 \rangle_w}{\text{var } r} + w_X \frac{\langle \Delta X^2 \rangle_w}{\text{var } X}$$

dimensionless;

$\langle \cdot \rangle_w$: average with a weight w_i per point.

Two questions:

1. **which weights** w_r, w_X ?
2. **which data set**?

set	points, weights
A	Zacharie-Aubrun + Onofri, standard UO_2 ; $w_i = 1$
B	all four papers; $w_i = 1$
C	as B, $w_i = \text{rank factor} \times \text{relevance}/3$

rank factor: A, B $\rightarrow 1$; $\text{U}^* \rightarrow 0.5$; $\text{U} \rightarrow 0.25$.

Where the data come from: quality rank $\times$ relevance

$$w_i = \underbrace{f(\text{Rose rank})}_{\text{how well it was measured}} \times \underbrace{\text{relevance}/3}_{\text{how close to our application}}$$

1. Rose rank — M.A. Rose, ANL/CFCT-22/26 (2022) (MSTDB-TP), re-worded for post-irradiation microscopy. Six aspects marked H/M/I: *method, calibration, sample characterisation, irradiation conditions, measurement precision, verifiability*.

rank	assigned when	f
A	all H/M <i>and</i> verifiability H	1
B	≥ 2 H/M <i>and</i> verifiab. H/M	1
U*	all but verifiability H/M, unc. unusable	0.5
U	uncertainties bar quantitative use	0.25

2. Relevance (1–3).

sample class	relevance
PWR standard UO ₂	3
PWR Cr-doped UO ₂	2
Halden disc, std	2
Halden disc, large grain	1

The calibrated parameters

Set C, $w_r = 0.2$, $w_X = 1$

all four papers, each point weighted by
quality rank $\times$ relevance

n (not fitted)	2
β	21.368
k	0.6788
ρ_{crit}	$6.854\text{e}14 \text{ m}^{-2}$

In sample, all four papers

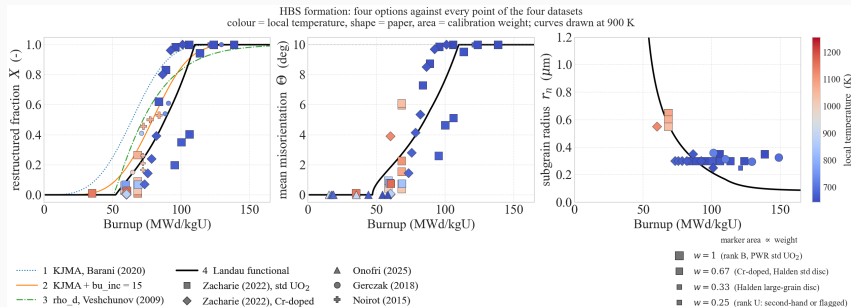
weighted, with the unweighted beside it

	N	RMSE / R^2	plain
Θ	52	$1.812^\circ / 0.756$	$1.778 / 0.772$
X	49	$0.188 / 0.740$	$0.177 / 0.763$
r_n	26	$0.135 \mu\text{m} / -0.21$	$0.134 / -0.26$

- X is in the objective.
- The size is still the price: $R^2 < 0$ on all 26 radii — standard UO_2 alone is $+0.30$, dragged down by the Cr-doped (-1.27) and Gerczak.
- $\rho_{\text{crit}} = 6.85 \times 10^{14} \text{ m}^{-2}$ is Eq. (1) at 47 GWd/tU: it is not an independent confirmation of the Veshchunov & Shestak range.

Results

Comparison with the other options and with all the data



Restructured fraction X , all $N = 49$ (weighted; plain in grey)

option	RMSE	R^2
1 KJMA	0.389	-0.11
2 KJMA + bu_{inc}	0.217	+0.65
3 ρ_d	0.232	+0.60
4 Landau	0.188	+0.74

plain: 0.380 / 0.204 / 0.227 / 0.177

Option 4 only: Θ 1.81° / +0.76
($N = 52$), r_n 0.135 μm / -0.21
($N = 26$). Fit C, the same weights.

Against the three models SCIANTIX already had

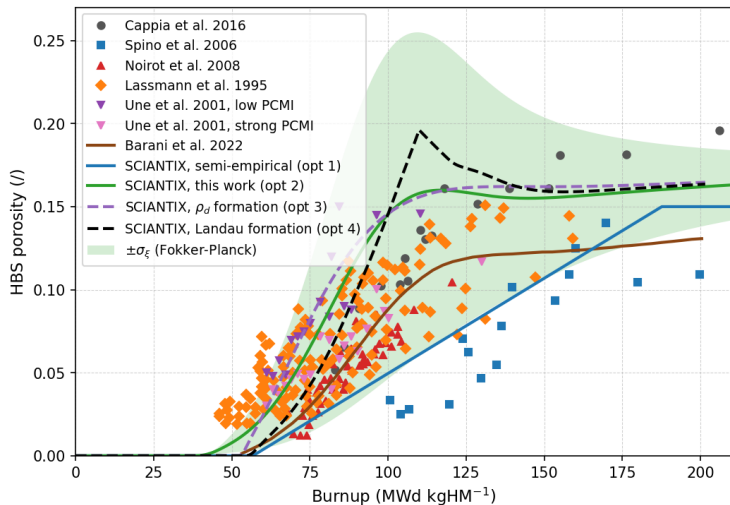
Burnup [MWd/kgU] at which α_r first reaches:

threshold	1 KJMA	2 KJMA+ bu_{inc}	3 ρ_d	4 Landau
1 %	19.5	34.5	51.2	53.1
10 %	37.8	52.8	54.8	60.9
50 %	64.2	79.3	72.3	88.3
90 %	90.1	105.2	112.3	106.5
99 %	109.7	124.6	161.7	109.7

- Option 4 starts **next to option 3** (53 against 51 MWd/kgU: both rest on a critical dislocation density) but rises more slowly: its 50 % crossing is the **latest of the four**.
- It still **saturates**: Θ hits the 10° cap and $X = 1$ at 110 MWd/kgU, whereas KJMA approaches 1 asymptotically.
- The 1–99 % span is **57 MWd/kgU**, against 90 for option 1 and 110 for option 3.

Model applied to SCIANTIX

Does the new formation model disturb the porosity model?



Option 4 overshoots to 0.178 at 110 MWd/kgU — its saturation — and dips by **13.6 %** after it. All four options still end within **0.36 %** of the same porosity.

Outlook: from a local
functional to a phase field

What a local functional cannot give -> phase-field models

Problems of the Landau-like local functional

- no spatial quantities;
- no kinetics: equilibrium is instantaneous.

family	property	application/ref
KWC	two fields (ϕ, θ), allows subgrain rotation; singular diffusion equations	Nucleation via subgrain coalescence, Muramatsu (2014)
MPF	One order parameter per grain; stable time increments; rotation ignored	Subgrain growth, Khajezade <i>et al.</i> (2024)
HMP	Removes nonphysical linear gradient term; handles inclination-dependent GB energy	Coupled Cosserat crystal plasticity, Tandogan <i>et al.</i> (2025); dislocation-driven nucleation, Tandogan <i>et al.</i> (2026)
PFC	Atomic-resolution density field; diffusive timescales	Interface morphology development during solidification, Nit-tala (2025)

Nucleation is usually **seeded ad hoc** at a critical density or stress; spontaneous nucleation from lattice-curvature gradients is obtained when coupling KWC with Cosserat crystal plasticity (Ghiglione, 2024), and from stored dislocations at grain boundaries when coupling HMP with it (Tandogan, 2026).

Muramatsu *et al.* (2014): a KWC phase field for recrystallization

The functional⁷ — two fields, ϕ and θ :

$$\mathcal{F} = \int_V \left[f(\phi) + \frac{\alpha^2}{2} |\nabla \phi|^2 + s g(\phi) |\nabla \theta| + \frac{\epsilon^2}{2} h(\phi) |\nabla \theta|^2 \right] dV$$

with $g(\phi) = h(\phi) = \phi^2$, relaxed by $\dot{\phi} = -M_\phi \delta \mathcal{F} / \delta \phi$ and $\dot{\theta} = -M_\theta \delta \mathcal{F} / \delta \theta$.

$\phi = 1$ inside a subgrain; θ is the **lattice orientation**.

The *singular* $|\nabla \theta|$ term gives a boundary an energy growing with misorientation.

The local part (Eq. 7, $f_m = 0$):

$$f(\phi; E_s) = a_{ub}(1 - p(\phi)) E_s + a_{wg} W_b q(\phi)$$

$$q(\phi) = \phi^2(1 - \phi)^2$$

$$p(\phi) = \phi^3(10 - 15\phi + 6\phi^2)$$

- q : the **double well**, original matrix against new grains; its height $a_{wg} W_b$ sets the boundary energy;
- p : is a sigmoid-like function where $p'(0) = p'(1) = 0$ therefore it *tilts* the well without moving its two minima;
- E_s : the **stored dislocation energy**, the only driving force, supplied from outside the model.

How the paper nucleates: a subgrain group enclosed by high-angle ($> 10^\circ$) boundaries is *selected* as a nucleus candidate and grows after an *incubation period* — the criterion is **imposed**, not produced by the free energy.

Tandogan *et al.* (2026): nucleation *from* the free energy

- Recrystallization is usually modelled with an imposed nucleation (Muramatsu);
- Tandogan *et al.*⁸ obtain it spontaneously, from stored dislocations at grain boundaries through a one thermodynamically consistent field theory, Cosserat crystal plasticity + Henry–Mellenthin–Plapp (HMP) orientation phase field (HMP-CCP).

Order parameters

- $\eta_T \in [0, 1]$: coarse-grained **crystalline order**, = 1 in the grain bulk, < 1 inside the diffuse GB.
- θ : lattice orientation, identified with the Cosserat microrotation
- 2D, small strain: dofs $(\eta_T, \theta, u_1, u_2)$, FEniCS + Mfront.

⁸Tandogan, Budnitzki, Sandfeld, *J. Mech. Phys. Solids* 206 (2026) 106325. Presented here up to §3.2.

The HMP-CCP free energy and the coupling function $\phi(\eta_T)$

$$\psi = f_0 \left[\alpha V(\eta_T) + \frac{\nu^2}{2} |\nabla \eta_T|^2 + \mu^2 g(\eta_T) |\nabla \theta|^2 \right] + \frac{1}{2} \mathbf{e}^e : \mathbb{E} : \mathbf{e}^e + 2\mu_c (\bar{\mathbf{e}}^e)^2 + \underbrace{\phi(\eta_T) \sum_{\alpha} \frac{\lambda}{2} \mu b^2 \rho_{\alpha}}_{\text{stored dislocations}}$$

- $V = \frac{1}{2}(1 - \eta_T)^2$: single well, penalises GBs.
- $g(\eta_T)$ singular at 1, with a $\ln$ term $\Rightarrow$ **Read-Shockley** GB energy.
- $\phi, \eta = \frac{c_3}{2} [1 - \tanh(c_1(\eta_T - c_2))]$: a **step**. The stored energy pushes on η_T only inside the GB ($\eta_T < c_2$); the bulk is “crystalline enough”.
- Recovery (Kocks–Mecking): extra sink $-\rho C_D A(\eta_T) \dot{\eta}_T$, active only where $\dot{\eta}_T > 0$: **behind a moving GB**.

$$\begin{aligned}
 (32) \quad & \tau_\eta \dot{\eta} = f_0 \nu^2 \nabla^2 \eta - f_0 [\alpha V' + \mu^2 g' |\nabla \theta|^2] - \phi' \sum_\alpha \frac{\lambda}{2} \mu_e r_\alpha^2 \quad (r_\alpha^2 \rightarrow b^2 \rho_\alpha) \\
 (33,34) \quad & 0 = f_0 \nabla \cdot [\mu^2 g(\eta) \nabla \theta] + 2\mu_e e^e, \quad -\tau_* \dot{e}_* = f_0 \nabla \cdot [\mu^2 g \nabla \theta], \quad \tau_* = \hat{\tau}_* g(\eta) \quad (23) \\
 (30,31) \quad & \Omega_\eta = \frac{1}{2} \tau_\eta^{-1} (\pi_\eta^{dis})^2, \quad \pi_\eta^{dis} = -\tau_\eta \dot{\eta} \\
 (35,36) \quad & V = \frac{1}{2} (1 - \eta)^2, \quad g = \frac{7\eta^3 - 6\eta^4}{(1 - \eta)^3} + c \ln(1 - \eta) + C_0, \quad C_0 = \min_{\eta_*} [\dots] + 0.01, \quad g(\min(\eta, 1 - 10^{-4})) \\
 (37,38) \quad & \phi(\eta) = \frac{1}{2} c_3 \{ \eta - c_1^{-1} \ln \cosh[c_1(c_2 - \eta)] \} + c_0, \quad \phi'(\eta) = \frac{c_3}{2} [1 - \tanh(c_1(\eta - c_2))] \\
 (28,29) \quad & \dot{\rho}_\alpha = \frac{1}{b} (K \sqrt{\sum_\beta a_{\alpha\beta} \rho_\beta} - 2d\rho_\alpha) |\dot{\gamma}_\alpha| - \rho_\alpha C_D A(\eta) \dot{\eta} \quad (\dot{\eta} > 0), \quad A = \frac{7\eta^3 - 6\eta^4}{1 - \eta} \\
 (41) \quad & f_{\eta 1} = f_0 \nu^2 \nabla^2 \eta, \quad f_{\eta 2} = -f_0 \alpha V', \quad f_{\eta 3} = -f_0 \mu^2 g' |\nabla \theta|^2, \quad f_{\eta 4} = -\phi' \frac{\lambda}{2} \mu_e b^2 \rho \\
 (42) \quad & \eta_{eq} = 1 - \frac{\phi'(\eta_{eq}) \frac{\lambda}{2} \mu_e b^2 \rho}{f_0 \alpha}, \quad \phi'(\eta_{eq}) = c_3 \text{ inside the widened GB}
 \end{aligned}$$

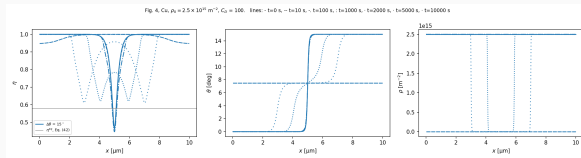
Table 1 (Cu): $f_0 = 371 \text{ kPa}$, $\alpha = 20$, $\nu = 1 \text{ } \mu\text{m}$, $\mu = 2.5/\pi \text{ } \mu\text{m}$, $c = 3$, $\lambda = 0.3$, $b = 0.2556 \text{ nm}$, $C_D = 100$, $c_1/c_2/c_3 = 100/0.95/1.7$.

The nucleation mechanism: a Cu bicrystal

Periodic bicrystal, $\Delta\theta = 15^\circ$, uniform $\rho_0 = 2.5 \times 10^{15} \text{ m}^{-2}$ switched on at $t = 0$.

$C_D = 0$, **no recovery**: the stored-energy force widens the GB into a non-localised orientation gradient, an *unstable* state.

$C_D > 0$: recovery at the GB centre: a dislocation-free nucleus bounded by two GBs that move outward.



Tandogan *et al.*, Fig. 4: η_T , θ , ρ and the forces on η_T ; red $C_D = 0$, blue $C_D = 100$.

Prototype: Tandogan in 1D, driven by the Landau model

What is solved — one GB at $x = L/2$, $L = 10 \mu\text{m}$, no load ($u = 0$, no slip, $\dot{\epsilon}^e = 0$):

$$\tau_\eta \dot{\eta}_T = f_0 \nu^2 \eta_T'' - f_0 [\alpha V' + \mu^2 g' \theta'^2] - \phi_{,\eta} \frac{\lambda}{2} G b^2 \rho$$

$$\hat{\tau}_* g \dot{\theta} = f_0 (\mu^2 g \theta')'$$

$$\dot{\rho} = \textcolor{brown}{S}(\textcolor{brown}{b}u) - \rho C_D A(\eta_T) \langle \dot{\eta}_T \rangle$$

- **Irradiation source** instead of plasticity: $S = \frac{d\rho_{\text{free}}}{d\textcolor{brown}{b}u} \dot{\textcolor{brown}{b}u}$, uniform, independent of the local $\rho \Rightarrow$ a recovered region is loaded again.

From the Landau model:

- $\Delta\theta \leftarrow \Theta(\textcolor{brown}{b}u)$, $\rho \leftarrow \rho_{\text{free}}(\textcolor{brown}{b}u)$ (the random array = Tandogan's SSDs), $G(T)$, b_{UO_2} ;
- other parameters as in the paper

Prototype: what comes out

figure tandogan_uo2_irradiation_free.png not generated

Open issues for discussion

1. The sweep: linear $\Delta R/R$ or volume?

What changed

- first version: $\rho_{\text{swept}} = (\rho_{\text{tot}} - \rho_{\text{ord}}) \Delta R/R$,
 $r_n \propto 1 + \Delta R/R$ — linear in the sweep;
- now: Gourdet–Montheillet volume,
 $x = k\rho_{\text{ord}}/\rho_{\text{tot}}$, $\rho_{\text{swept}} \propto 1 - e^{-x}$,
 $S/V \rightarrow S/V e^{-x}$.

It matters: $x \approx 0.5$ at 60 GWd/tU, $e^{-x} = 0.6$
against $1 - x = 0.5$.

Geometry: a shell $\Delta R/R$ sweeps
 $1 - (1 - \Delta R/R)^3 \approx 3 \Delta R/R$ of the volume; radius
as function of volume goes as $(1 + x)^{1/3}$.

Literature

- Gourdet & Montheillet: dV is a swept volume.
- Rest (2004–06): linear annulus depth
 $V_r = 1 - [1 - r/(d_g/2)]^3$, and the lattice
strain $\Delta a/a_0$ as driving quantity.

Questions

- is $k\rho_{\text{ord}}/\rho_{\text{tot}}$ a length or a volume fraction?
- should the swept walls also leave the
energy ($\rho_{\text{ord}} \rightarrow \rho_{\text{ord}} e^{-x}$ in F)?
- sensitivity of Θ , X , r_n to the choice: one
refit with the linear form.

2. Dataset differences: irradiation conditions and ranking

dataset	technique	conditions	observable
Zacharie-Aubrun 2022	EBSD, PWR	local burnup (model), grains 9–10 μm std, 56–60 μm Cr	f_1 , f_{10} , AMis, ECD
Onofri 2025	EBSD + TEM, 16 and 37 GWd/t	model burnup, grain size default	AMis, no f_1
Gerczak 2018	SEM-BSE, EBSD, tKD, H.B. Robinson	burnup from literature (Barani)	rim d_A , area (rank U)
Noirot 2015	EPMA, Halden discs	enrichment 19.8 %, fission rate $\sim 2\times$ PWR	Xe-depleted area

- Each dataset has **its own irradiation conditions**, and is already ranked differently:
 $w = f(\text{Rose rank}) \times \text{relevance}/3$ (Zacharie std carries about 70 % of the fraction weight).
- Different observables are fitted as one “fraction” (EBSD f_{10} , Xe-depleted area, BSE image analysis); Θ is T -independent while centre points ($\sim 900^\circ\text{C}$) and rim points ($\sim 400^\circ\text{C}$) sit at the same burnup.
- At 35 GWd/t: AMis $0.4\text{--}0.7^\circ$ (Onofri) against 0.9° (Zacharie); the model gives 0 for both.

To discuss: is rank $\times$ relevance enough, or should the weight depend on T , fission rate, enrichment (a condition-aware rank, or a per-dataset onset offset)? Leave-one-dataset-out fits show the sensitivity. And: use the *previous formation curves* (options 1–3) as extra, low-weight calibration targets for $X(bu)$? Risk: circular, they are fits of the same biased data.

3. Critical densities

source	value
this model, ρ_{crit}	$6.85 \times 10^{14} \text{ m}^{-2}$ (47.1 GWd/tU on the Nogita line)
Nogita & Une (1994), measured	5.8×10^{14} at 44, 6.0×10^{14} at 83 GWd/t
Veshchunov & Shestak (2009)	6×10^{14} to $\sim 10^{15} \text{ m}^{-2}$
Matzke & Kinoshita (1997)	~ 70 GWd/t local
Sontheimer (2011)	75 GWd/tU
model: $X > 0$	52.3 GWd/tU

- The Veshchunov value is **not independent**: it is Nogita's datum at 44 GWd/t, and ρ_{crit} is Eq. (1) at 47.
- Eq. (1) extrapolated to 83 GWd/t gives 4.2×10^{15} against 6.0×10^{14} measured.
- Onofri's TEM line density, 1.1×10^{13} and $0.8 \times 10^{13} \text{ m}^{-2}$ at 16 and 37 GWd/t, is 10–50 $\times$ below Eq. (1) and nearly burnup-independent.
- A_1 uses the excess density $\rho_N - \rho_{\text{crit}}$ (3.5×10^{12} at 47.2 GWd/tU), not the real one ($\sim 7 \times 10^{14}$).

4. Phase-field functions for the HBS formation, and the problems met

	Muramatsu (KWC)	Tandogan (HMP-CCP)	Landau (here)
order parameter	ϕ nucleus/matrix, θ	η crystal/boundary, θ	$\eta = \theta/\theta_{\max}$, mean field
role of ρ	$E_s = \rho \mu b^2/2$ tilts $f(\phi)$	$\phi(\eta)\rho b^2$ tilts η in the GB only	ρ_{tot} split in 3 populations
threshold	$r^* = 2\gamma_b/E_s$, criteria (11),(13), imposed	$\eta_{\text{eq}} < 1$, then recovery $A(\eta)$	ρ_{crit}

Problems of the 1D UO₂ prototype

- *width*: $\nu = 2.5 \mu\text{m}$ (GB width $\sim 5 \mu\text{m}$) against subgrains of $0.2\text{--}0.8 \mu\text{m}$: the smallest feature is not resolved;
- *onset is an input*: $c_3 = 2.659$ is fitted to nucleate at 60 GWd/tU; $\eta_{\text{eq}} < 0$ beyond ~ 81 GWd/tU;
- Cu parameters, placeholder time scale, $C_D = 10$ instead of 100, no thermal activation;
- *nothing makes misorientation*: η, θ redistribute the parent angles, so $\Theta(bu)$ is not reproduced; 1D has no curvature;
- archived prototype: $\nu = 50 \text{ nm}$, $f_0 = 10.8 \text{ MPa}$ and a Read–Shockley target $\sim 35\%$ above Zhang (1.59 against 1.18 J/m^2 at 10°).

Which free-energy function should carry ρ , and where does the threshold come from?

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Backup — the equations

$$(1) \rho_{\text{tot}}(bu) = \max(10^{2.2 \times 10^{-2} bu + 13.8} - \rho_{\text{crit}}, 0)$$

$$(2) G(T, P, x, q), \nu(T, P, x, q) \quad \text{NEA/NSC/R(2024)1 p. 124; } f(\nu) = \frac{1-\nu/2}{1-\nu}$$

$$(3) \rho_{\text{LAGB}}^{\text{max}} = \left(\frac{3n\theta_{\text{max}}}{\beta b} \right)^2, \quad (S/V)_{\text{max}} = \frac{9n\theta_{\text{max}}}{\beta^2 b}, \quad x_{\text{max}} = \frac{k \rho_{\text{LAGB}}^{\text{max}}}{\rho_{\text{tot}}}$$

$$(4) A_1 = \frac{f(\nu)}{4\pi} \ln \frac{\rho_{\text{tot}}^{-1/2}}{b}, \quad A_2 = \frac{f(\nu)}{4\pi} \ln \frac{\min(b/\theta, \rho_{\text{tot}}^{-1/2})}{b}$$

$$(5) \rho_{\text{ord}} = \rho_{\text{LAGB}}^{\text{max}} \eta^2, \quad x = x_{\text{max}} \eta^2, \quad \rho_{\text{free}} = (\rho_{\text{tot}} - \rho_{\text{ord}}) e^{-x}, \quad \rho_{\text{swept}} = (\rho_{\text{tot}} - \rho_{\text{ord}})(1 - e^{-x})$$

$$(6) F = \rho_{\text{free}} A_1 G b^2 + \rho_{\text{ord}} A_2 G b^2 = C_0 + E_{\text{wall}} + E_{\text{sweep}}$$

$$(7,7b) \eta_{\text{eq}} = \text{argmin}_{[0, \min(\eta_{\text{bal}}, 1)]} F, \quad \eta_{\text{bal}} = \sqrt{\min(\rho_{\text{tot}} / \rho_{\text{LAGB}}^{\text{max}}, 1)}$$

$$(8) \Theta = \min(\eta_{\text{eq}} \theta_{\text{max}} \cdot \frac{180}{\pi}, \theta_{\text{HAGB}})$$

$$(9) r_n = \min\left(\frac{1.5}{(S/V)_{\text{max}} \eta e^{-x}}, R_{\text{grain}}\right)$$

$$(10) X = \text{clip}\left(\frac{\Theta - \theta_u}{\theta_{\text{HAGB}} - \theta_u}, 0, 1 - 10^{-9}\right)$$

Backup — regimes and parameters

Regimes (T -, P -, x -independent)	
condition	b_u [GWd/tU]
$\rho_{\text{tot}} > 0$, Θ leaves 0	47.09
$\Theta = \theta_u$, X leaves 0	52.27
$\Theta = \theta_{\text{HAGB}}$	109.96

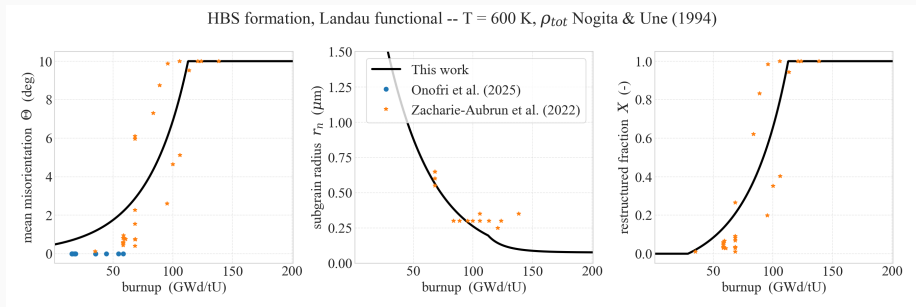
With the shipped set-C parameters the balance bound of Eq. (7b) **binds from 89 to 110 GWd/tU**; below it η is the interior minimum of Eq. (7).

Fixed constants

b	3.889e-10 m
θ_{max}	$= \theta_{\text{HAGB}}$
θ_{HAGB}	10°
θ_u	1.00°
n	2

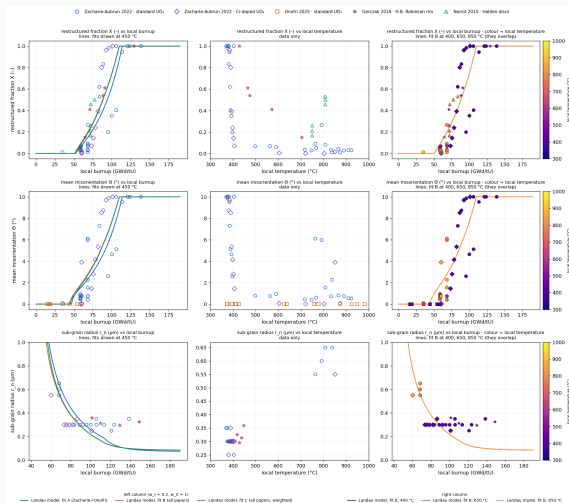
θ_u is the lower end member of the mixture Eq. (10) inverts, and it is set by the **measurement convention**, not fitted: the EBSD data reports a restructured fraction *at* 1° (f_1) and one *at* 10° (f_{10}), so 1° is the edge below which a boundary is not counted at all. A map whose whole resolved population sits there carries no restructuring — which is $X = 0$. It fixes only *where* X leaves zero: Θ and r_n do not depend on it.

Backup — the model on its own, $T = 600$ K, $P = 0.05$



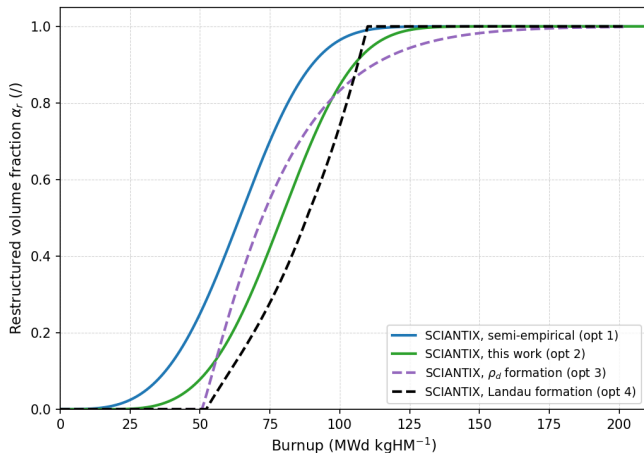
```
python3 hbs_formation_landau.py --plot
```

Backup — fits A, B, C against the data



```
python3 calibrate.py --study --weight 0.2 --fraction-weight 1
```

Backup — the four regression cases, as they ship

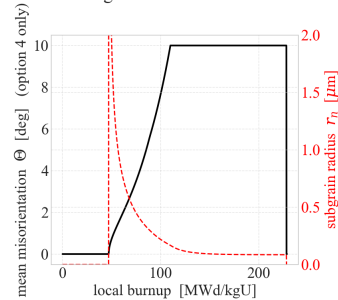
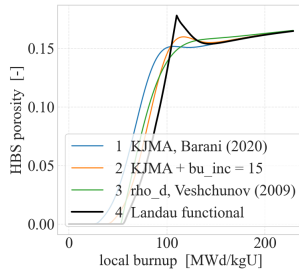
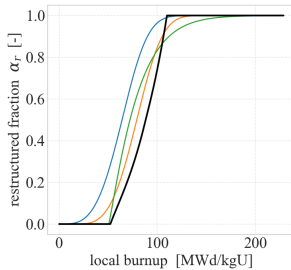


Not the controlled comparison of the previous figure: each case pairs its formation option with the porosity model it was built around (1, 2, 3, 3), so part of the spread is the porosity model.

This is what a user of each regression case actually gets;
`formation_options.png` is what isolates the formation model.

Backup — the four formation options in a full run

HBS formation options on the same irradiation, porosity model 3 throughout



Backup — the 1D HMP prototype: implemented equations

Fields $\eta_{\text{T}}(x, t)$, $\theta(x, t)$, $\rho(x, t)$ on $x \in [0, L]$, one GB at $L/2$;
 $u = 0$, no slip, $\theta = -e^*$ (Eq. 27)

$$(32) \quad \tau_{\eta} \dot{\eta}_{\text{T}} = f_0 \nu^2 \eta_{\text{T}}'' - f_0 [\alpha V'(\eta_{\text{T}}) + \mu^2 g'(\eta_{\text{T}}) \theta'^2] - \phi'(\eta_{\text{T}}) \frac{\lambda}{2} G b^2 \rho$$

$$(34) \quad \hat{\tau} g(\eta_{\text{T}}) \dot{\theta} = f_0 (\mu^2 g(\eta_{\text{T}}) \theta')'$$

$$(28) \quad \dot{\rho} = S(bu) - \rho C_D A(\eta_{\text{T}}) \langle \dot{\eta}_{\text{T}} \rangle, \quad \langle \cdot \rangle = \max(\cdot, 0)$$

$$V = \frac{1}{2}(1 - \eta_{\text{T}})^2, \quad A = \frac{7\eta_{\text{T}}^3 - 6\eta_{\text{T}}^4}{1 - \eta_{\text{T}}}$$

$$g = \frac{7\eta_{\text{T}}^3 - 6\eta_{\text{T}}^4}{(1 - \eta_{\text{T}})^3} + c \ln(1 - \eta_{\text{T}}) + g_{\text{off}}, \quad \min g = 0.01$$

$$\phi' = \frac{c_3}{2} [1 - \tanh(c_1(\eta_{\text{T}} - c_2))]$$

$$(42) \quad \eta_{\text{T}}^{\text{eq}} = 1 - \frac{c_3 \frac{\lambda}{2} G b^2 \rho}{f_0 \alpha}$$

g , g' , A frozen for $\eta_{\text{T}} > 1 - 10^{-4}$; $\eta'_{\text{T}} = \theta' = 0$ at $x = 0, L$.

Coupling to the Landau model, `hbs_state(bu, T)`

$$\Delta\theta = \Theta(bu), \quad \rho \leftarrow \rho_{\text{free}}(bu), \quad G = G(T, \dots)$$

$$c_3 = \frac{(1 - 0.579) f_0 \alpha}{\frac{\lambda}{2} G b^2 \rho_{\text{free}}(bu_{\text{sat}})}$$

so that Eq. (42) gives the paper's Cu value 0.579
 where $\Theta = \theta_{\text{HAGB}}$.

Two protocols for ρ

- *step*: $S = 0$, $\rho(x, 0) = \rho_{\text{free}}(bu)$ (as in §3.2);
- *irradiation*: $\rho(x, 0) = 0$, $S = \frac{d\rho_{\text{ref}}}{d bu} \dot{bu}$,
 uniform in x , $\rho_{\text{ref}} \in \{\rho_{\text{free}}, \rho_{\text{tot}}\}$,
 $bu = bu_0 + \dot{bu} t$ (compressed).

Nucleus: $\rho(L/2) < 0.1 \rho_0$ while $\min \eta_{\text{T}}(0, L) > c_2$.

Capillarity (absent in 1D): $r^* = 2\gamma / (\frac{\lambda}{2} G b^2 \rho)$.

Interface parameters: Cu, Table 1

$(f_0, \nu, \alpha, \mu, c, \tau_{\eta}, \hat{\tau}, C_D)$, not calibrated for UO_2 .

FV, 400 cells, BDF.